Hybrid Classical–Neural Detection of Voice Pathology from Sustained Vowels with Cross-Database Evaluation

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***Abstract*—We propose a non-invasive screening system that labels a short sustained-vowel recording as *healthy* or *pathological*. The system combines hand-implemented classical speech analysis—the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCC), linear predictive coding (LPC), and pitch-perturbation measures—with a lightweight convolutional neural network (CNN) in a late-fusion architecture. An MFCC–Gaussian mixture model (GMM) classifier serves as the baseline. The system is trained on the public Saarbrücken Voice Database (SVD) under a speaker-independent protocol and tested, without retraining, on the independent VOICED database recorded with different equipment.**

***Index Terms*—voice pathology detection, MFCC, linear prediction, jitter and shimmer, CNN, cross-database evaluation**

I. Problem and Motivation

Pathologies such as nodules, polyps, laryngitis, and vocal-fold paralysis change how the vocal folds vibrate. The result is a voice that sounds hoarse, breathy, or rough. A clinical examination is the reference standard, but it requires an otolaryngologist and dedicated equipment. A recording-based screening tool could flag at-risk speakers early and support remote follow-up. It is a decision-support aid, not a replacement for diagnosis.

Many studies report high accuracy on the SVD. However, Huckvale and Buciuleac [1] showed that these numbers are hard to compare: papers use different pathology subsets, audio materials, and cross-validation schemes. When three published systems were re-implemented under one protocol, the differences between them largely disappeared. End-to-end deep learning on small clinical data is not automatically better either: a convolutional–recurrent network trained on raw SVD vowels reached 68% accuracy [2].

Most of these disorders alter how regularly the vocal folds vibrate, while the resonances of the vocal tract change comparatively little. Cepstral coefficients such as MFCCs [4] summarize mainly those resonances [3]. This project therefore pairs them with measurements taken directly from the excitation—the LPC prediction residual and the cycle-to-cycle variation of pitch period and amplitude (jitter, shimmer)—and with a spectrogram-based CNN, and compares all of them on the same folds.

II. Research Questions

The project is built around three questions: *Q1 (accuracy).* Does the fused model reach a higher UAR than an MFCC–GMM classifier when no speaker appears in both the training and the test data? *Q2 (robustness).* When a model trained on SVD is applied to VOICED, which input—cepstral statistics, excitation measures, or the spectrogram—loses the least performance? *Q3 (fairness).* Are the errors evenly distributed between female and male speakers?

III. Data

SVD [5], [6]: German speakers recorded at 50 kHz, 16-bit. It contains 1002 speakers with a wide range of voice disorders and 851 controls [1]. Each session has /a/, /i/, /u/ at normal, high, low, and rising–falling pitch, plus one sentence. The database is released on Zenodo under CC BY 4.0 (DOI 10.5281/zenodo.16874898). Of the 38 GB release, only the normal-pitch /a/ recordings are used, converted from `.nsp` to `.wav` with the published converter.

VOICED [7]: 208 recordings of a 5 s sustained /a/, sampled at 8 kHz and released on PhysioNet under the ODC Attribution License. It is used *only* as an external test set.

TABLE I
Databases (SVD: speakers in the full database [1]).

Database	Healthy / Path.	Fs	Role
SVD	851 / 1002 spk.	50 kHz	train / CV
VOICED	58 / 150 rec.	8 kHz	external test

Task: binary classification, healthy vs. any pathology. Sessions from the same speaker are always kept in the same fold to prevent speaker leakage (some SVD speakers have many sessions [1]). For the cross-database experiment, the SVD is downsampled to 8 kHz with an anti-aliasing filter so that the training and test bandwidths match.

IV. Methodology

Every signal-analysis block is coded by hand with NumPy, without feature-extraction libraries, and checked by unit tests

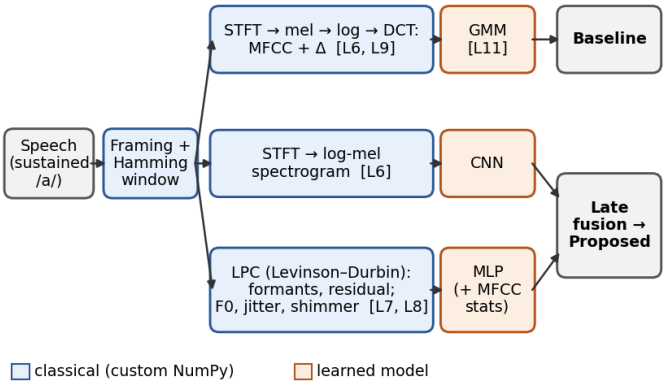


Fig. 1. Pipeline. Blue: signal analysis coded by hand (lecture numbers in brackets). Orange: trained models.

against a reference implementation. The lecture in which each method was covered is given in parentheses.

- *Spectral analysis* (L6, L9): pre-emphasis, 25 ms Hamming-windowed frames with a 10 ms hop, and an FFT, followed by a 40-band mel filterbank, logarithm, and DCT-II. This gives 13 MFCCs plus Δ coefficients; the same filterbank output without the DCT is the log-mel spectrogram used by the CNN.
- *Linear prediction* (L7): order-16 LPC with the autocorrelation method and the Levinson–Durbin recursion [8]. From it, the first three formants and their bandwidths (polynomial roots) are derived. Inverse filtering gives the prediction residual $e[n] = s[n] - \sum_{k=1}^{p} a_k\, s[n-k]$, an estimate of the excitation. Its normalized energy and kurtosis are computed.
- *Pitch and perturbation* (L8): F0 is estimated by autocorrelation. Local jitter and shimmer are computed from consecutive period lengths T_i and peak amplitudes A_i; for example, jitter $= \frac{1}{N-1}\sum_i |T_i - T_{i+1}| \,/\, \bar{T}$.

Frame-level values are reduced to one vector per recording by their mean and standard deviation.

Reference system. One diagonal-covariance GMM per class [9] is trained on frame-level MFCC+Δ vectors. A recording is assigned to the class with the larger total log-likelihood over its frames; the number of mixture components is tuned by inner cross-validation.

Fused model (Fig. 1). A small CNN—three convolution, batch-normalization and pooling stages followed by global average pooling—reads a fixed-length log-mel excerpt. In parallel, a two-layer MLP reads the standardized vector of cepstral statistics, LPC values and perturbation measures. The two embeddings are concatenated and mapped to a single sigmoid output. Training uses class-weighted binary cross-entropy with early stopping. To see where any gain comes from, the model is also retrained with one input branch removed at a time.

Prediction. Since the disorders act mainly on vocal-fold vibration, we expect the excitation measures to degrade least when the microphone and sampling rate change (Q2), and the spectrogram branch to give the best accuracy within SVD (Q1).

V. Evaluation and Plan

Experiments. E1 (Q1): stratified 5-fold cross-validation on SVD with speaker-disjoint folds, as recommended in [1]. E2 (Q2): every system is retrained on all SVD data at 8 kHz and scored on VOICED with no adaptation. E3 (Q3): E1 results are split by sex, and E1 is repeated with a sex-balanced training set. The branch-removal runs are reported for both E1 and E2.

Metrics. The primary score is the unweighted average recall (UAR), because the two classes are not the same size. F1, sensitivity, specificity and ROC AUC are also given, as mean ± standard deviation over the folds, and systems are compared with paired tests on the fold scores.

Implementation. Experiments run on Google Colab; audio subsets, cached `.npy` features and model checkpoints are kept on Google Drive. The code and the exact fold lists are published at https://github.com/mmerts/voice-pathology-detection.

Anticipated difficulties. The data set is small and unbalanced, so the CNN is kept compact and trained with class weights and early stopping. Pitch tracking can fail on strongly disordered voices; such frames are excluded from jitter and shimmer, and the share of excluded frames becomes an extra feature. The two databases differ in sampling rate, which is why E2 is run entirely at 8 kHz.

Schedule and deliverables. 20 Nov: data prepared and analysis blocks passing their tests. 4 Dec: GMM reference and fold framework. 18 Dec: fused model, branch-removal runs, E2 and E3. 25 Dec: final paper and code. 28 Dec–1 Jan: presentation. The final report will contain one table comparing the reference system, each single branch and the fused model under E1 and E2, which answers Q1–Q3 directly.

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